

Information Systems

Name: Allan Kipkorir Mutai

Course: Master of Information Technology

Date: 08/06/2025

Table of Contents

Information Systems	1
Introduction.....	3
Prompts	3
First Gemini Prompt: “Hey Gemini, how can information systems in organizations be used to perform everyday business activities in alignment with business strategies of e-Commerce systems such as Shopify, WooCommerce, Amazon, or others.” (Google Gemini, 2026- First prompt)	3
From Scholarly Sources:.....	3
AI Critique:	4
Last Gemini Prompt: “Hey Gemini, assist me in Criticising the ethical, social, privacy and security issues related to information systems used in e-Commerce systems such as Shopify, WooCommerce, Amazon, or others.” (Google Gemini, 2026- Second prompt).....	4
From Scholarly Sources	4
AI Critique	4
References	6

Introduction

This report aims to analyse Gemini's response and critique its responses comparing with scholarly sources and analysing if these information systems are as accurate as we expect them to be.

Prompts

First Gemini Prompt: "Hey Gemini, how can information systems in organizations be used to perform everyday business activities in alignment with business strategies of e-Commerce systems such as Shopify, WooCommerce, Amazon, or others." (Google Gemini, 2026- First prompt)

The prompt aims to use AI intelligence to analyse how different E-commerce systems are being used in the real world to align business activities to business strategies.

The output from the Gemini prompt above is a break down of the whole process into three distinctive core layers which are:

- a. Strategic Alignment: this layer guides the business strategy.
- b. Information System Hub: this layer is responsible for illustrating integration of core systems.
- c. Routine Business Activities: this part is responsible for translating systems into daily operations.

From Scholarly Sources:

Based on my reading of scholarly sources, the main takeaway is information systems are a combination of hardware, software, processes and data designed to manage information. For any organization to achieve success, the technological infrastructure must mirror and support its business objective. (Dairo et al., 2021)

Organizations execute a huge number of operations behind the scenes (Singh, 2021). The most important routine activities in an E-Commerce environment include:

- a. Stock Management – This involves the use of Information systems to track inventory in real time. Inventory management allows the business become accountable for its resources.
- b. Order and Payment Processing – Safe and secure transaction allow business run, e-commerce businesses take credit card information of the clients, verify funds, send order receipt and start order processing
- c. Consumer Data handling – This is a crucial factor which involves storing customer data, addresses, purchases and shopping carts
- d. Logistics and Tracking – this is a sensitive section of any e-commerce system that involves generating shipping labels, notification of carriers and updating customers of their goods in transit.

AI Critique:

- Is the information accurate and complete? YES, but lacks clarity and real-world examples.
- Are any important issues missing or oversimplified? The AI is giving more of a holistic overview.
- How does it compare with academic or industry sources? AI gives an educational overview while scholarly sources give us something based off of the real world.
- Would it be useful in a real business decision-making context? Yes, if you are looking to start any related business.
- Does it include bias or unsupported assumptions? Not that I have seen. All the points seem to match.

In my point of view, both Scholarly sources together with AI response kind of seem to be discussing the same concepts. The AI results seem to be a bit of a holistic approach while the scholarly sources seem to go quite in depth and deliver real-world scenarios that these E-commerce platforms must deal with.

Last Gemini Prompt: “Hey Gemini, assist me in Criticising the ethical, social, privacy and security issues related to information systems used in e-Commerce systems such as Shopify, WooCommerce, Amazon, or others.” (Google Gemini, 2026- Second prompt)

The expansion of e-commerce platforms like Amazon, Shopify, and WooCommerce has revolutionized global retail. However, the information systems (IS) powering these platforms—including Order Management Systems (OMS), Customer Relationship Management (CRM), and AI-driven analytics—create significant ethical, social, privacy, and security challenges (Gemini,2026).

From Scholarly Sources:

Information systems enable e-commerce to run smoothly and profitable but also pose important and very interconnected risks. Advanced technology for tracking, analysing and selling to humans creates a complex web of ethical, social, privacy and security dilemmas (PwC. (2025)).

Ethical issues deal with what is "right versus wrong" in how companies treat people, even if those actions are legally permitted (Stallion Journal, 2022)

The challenge of modern e-commerce goes beyond mere legal compliance with regulations like GDPR or CCPA, to adopting a more principled privacy-first strategy that respects users' boundaries, rather than treating human behaviour as a raw material to be harvested and sold (Harvard Commentary, 2026; Taylor & Francis, 2026).

AI Critique

- Is the information accurate and complete? yes

- Are any important issues missing or oversimplified? Not really, all the information seem to match
- How does it compare with academic or industry sources? They are kind of similar, talking about the same subject matter.
- Would it be useful in a real business decision-making context? I guess it would be important to be considerate of the risks associated with technology being introduced to E-commerce.
- Does it include bias or unsupported assumptions? No.

References

- Dairo, M., Adekola, J., Apostolopoulos, C., & Tsaramirsis, G. (2021). Benchmarking strategic alignment of business and IT strategies: opportunities, risks, challenges and solutions. *International Journal of Information Technology*, 13(6), 2191–2197. <https://doi.org/10.1007/s41870-021-00815-7>
- Singh, B. M. (2021). Challenges in adoption and implementation of E-Commerce platform: Case study of Nepal. *LBEF Research Journal*, 3(2), 56–68.
- Harvard Commentary. (2026). Defending Privacy in the Digital Age: Reflections for Data Privacy Day 2026. *Carr Center for Human Rights Policy at Harvard Kennedy School*. <https://www.hks.harvard.edu/centers/carr-ryan/our-work/carr-ryan-commentary/defending-privacy-digital-age-reflections-data>
- PwC. (2025). Global Digital Trust Insights: Ethical concerns around data privacy and personalization in digital marketing. *PwC / ResearchGate Publication*. <https://www.researchgate.net/publication/399114991>
- Stallion Journal. (2022). Security and Privacy Concerns in E-commerce: A Consumer Perspective. *Stallion Journal for Multidisciplinary Associated Research Studies*, 1(2), 99-107. <https://doi.org/10.55544/sjmars.1.2.13>
- Google Gemini, 2026. Gemini (Advanced version). [AI chat log]. Available at: <https://gemini.google.com/app/a2bbe21f9016d699> [Accessed 8 June 2026].